

VAKA HAREESH REDDY

B.Tech Computer Science Engineering (AI & ML)

8309628645 | hareeshreddyvaka2006@gmail.com | linkedin.com/in/hareesh-reddy-vaka | Chennai, Tamil Nadu, India

PROFESSIONAL SUMMARY

Third-year Computer Science student (AI/ML specialization) at VIT Chennai with a 7.74 CGPA. Over the past year, I have built and deployed projects across deep learning, multi-agent reinforcement learning, explainable AI, and applied cryptography mostly as graded coursework, some on my own time. I learn fastest by building things, which is why most of what I know came from projects rather than lectures. Looking for roles in AI/ML Engineering, Data Science, or Software Development where I can do the same.

EDUCATION

Vellore Institute of Technology (VIT), Chennai

Aug 2023 – Jul 2027

B.Tech in Computer Science Engineering Specialization: Artificial Intelligence & Machine Learning Chennai, Tamil Nadu

- CGPA: 7.74 | No active backlogs
- Relevant Coursework: Deep Learning, Reinforcement Learning, Explainable AI, Compiler Design, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, SQL, JavaScript, HTML, CSS, R

ML / DL Frameworks: TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, MediaPipe

AI / ML Concepts: Convolutional Neural Networks (CNN), LSTM/GRU, Transformer Architectures, Ensemble Learning, Multi-Agent Reinforcement Learning (MAPPO/PPO), Curriculum Learning, Generalized Advantage Estimation (GAE), Explainable AI (SHAP, Grad-CAM, Saliency Maps), Computer Vision

Web & Backend: Flask, FastAPI, React, Node.js, REST APIs

Data & Visualization: pandas, NumPy, Matplotlib, Seaborn

Security & Cryptography: AES-256-GCM, RSA-2048, TOTP 2FA, OAuth2

Developer Tools & Platforms: Git, GitHub, Docker, Google Cloud Platform, n8n (Workflow Automation), Gmail API, Google Gemini API, VS Code, PyCharm, Jupyter Notebook, Google Colab

Databases: MySQL, SQLite

ACADEMIC & PERSONAL PROJECTS

Face Shape Classification & Virtual Hairstyle Try-On System

Jan – May 2025

Python · TensorFlow · EfficientNet-B3 · MobileNetV2 · CoAtNet-Tiny · MediaPipe · OpenCV

- Built a three-model ensemble (EfficientNet-B3, MobileNetV2, CoAtNet-Tiny) using weighted soft-voting; hit ~89.6% test accuracy across five face shape categories.
- Built a virtual hairstyle try-on pipeline using MediaPipe facial landmarks and alpha compositing to overlay hairstyle images onto input photos.
- Used random flip, rotation, zoom, and brightness jitter during training to help the model generalize across different lighting and skin tones.
- Documented model architecture, evaluation metrics, and a comparison between individual model and ensemble performance.

Multi-Agent Warehouse Robot Coordination (MARL)

Jan – May 2025

Python · PyTorch · MAPPO · Curriculum Learning · GAE · Reinforcement Learning

- Built a MAPPO system with a centralized critic to coordinate multiple warehouse robots each agent acts independently but the critic sees the full state during training.
- Used curriculum learning across three grid sizes (10×10, 15×15, 20×20) with varying agent counts; starting small let agents learn basic coordination before the environment got harder.
- Applied Generalized Advantage Estimation (GAE) to reduce gradient variance, which made training noticeably more stable.
- Tracked task completion rate, collision count, path efficiency, and average episode reward at each curriculum stage to measure actual progress.

DeepHealthX Explainable Cardiac Disease Detection

Jan – May 2025

Python · TensorFlow · SHAP · Grad-CAM · Temporal Gradient Saliency · Multi-modal

- Built a three-input deep learning model that takes clinical lab data, ECG signals, and cardiac MRI scans processed separately, then fused to predict cardiac disease.
- Added an XAI layer: SHAP for feature attribution on tabular inputs, Grad-CAM for MRI spatial attention, and temporal gradient saliency to trace ECG-based decisions.
- The goal was to make the model's reasoning visible to a clinician, not just report a prediction which also made it easier to catch mistakes during development.
- Wrote up the work as an IEEE Access-format paper covering architecture, XAI approach, and results.

SecureDocs Cryptographic File Sharing Application

2025

Python · Flask · AES-256-GCM · RSA-2048 · TOTP 2FA · Audit Logging

- Built a Flask web app for secure file sharing: AES-256-GCM for file encryption, RSA-2048 for key exchange, and TOTP-based 2FA for login.
- Every file operation gets written to an audit log; document owners can grant or revoke recipient access with expiry controls.
- Added simulated cloud storage and KMS demo pages to show how the system would behave in an actual enterprise setup.
- The whole thing was built as a Compiler Design project which made the security architecture more deliberate than it might otherwise have been.

Automated Placement & Internship Opportunity Tracker

2025

n8n · Gmail API (OAuth2 Readonly) · Google Gemini · Workflow Automation

- Set up an n8n workflow that watches Gmail via OAuth2 and catches incoming placement and internship emails automatically no manual checking needed.
- Used Google Gemini to pull structured fields (company, role, deadline, CTC, eligibility) out of raw email text; the unstructured format of these emails made this harder than it sounds.
- Handled base64-encoded MIME payloads in the workflow and routed parsed data into a centralized tracker.

Car Rental Feedback Analyzer

2024

Python · IBM watsonx.ai · Sentiment Analysis

- Used IBM watsonx.ai to classify customer reviews as positive, neutral, or negative, then visualized the sentiment breakdown to find recurring complaint patterns.
- The main finding: negative reviews clustered around a few specific issues, which made it straightforward to flag actionable areas for improvement.

Responsive Restaurant Website

2023

HTML · CSS · JavaScript

- Built a multi-page restaurant site with table booking, a feedback form, and a contact page all responsive across screen sizes.
- Focused on mobile layout and basic accessibility; tested across common browsers.

CERTIFICATIONS & TRAINING

- IBM Generative AI Certification IBM SkillsBuild (2024)

TOOLS & TECHNOLOGIES AT A GLANCE

Category

Deep Learning

Computer Vision

Reinforcement Learning

Explainable AI

NLP

Automation

Security

Web Development

Cloud & DevOps

Tools

TensorFlow, PyTorch, Keras, EfficientNet, MobileNet, CoAtNet

OpenCV, MediaPipe, Grad-CAM, Alpha Compositing

MAPPO, PPO, GAE, Curriculum Learning (PyTorch)

SHAP, Grad-CAM, Temporal Gradient Saliency

IBM watsonx.ai, Google Gemini, Prompt Engineering

n8n, Gmail API, Google Gemini API, OAuth2

AES-256-GCM, RSA-2048, TOTP, Flask

React, Node.js, FastAPI, Flask, HTML/CSS/JS

Google Cloud Platform, Docker, GitHub

ACADEMIC HIGHLIGHTS

- 7.74 CGPA at VIT Chennai with AI/ML specialization; no backlogs.
- Five project-based assessments completed in Deep Learning, Reinforcement Learning, Explainable AI, and Compiler Design all individual or small-team builds.
- Self-taught workflow automation (n8n), API integration (Gmail, Gemini), and applied cryptography outside of coursework, mostly to solve real problems.
- 120+ LeetCode problems solved, mostly arrays, graphs, and dynamic programming still a work in progress.

INTERESTS & AREAS OF EXPLORATION

- AI/ML particularly multi-agent systems, generative models, and making models explainable.
- Workflow automation and LLM integration wiring APIs and models together to automate repetitive tasks.
- Applied cryptography and secure system design.
- Competitive programming and algorithm problems slower going than I'd like, but improving.

LANGUAGES

- Telugu Native
- English Professional Proficiency
- Hindi Conversational